18-Week Combined Data Engineering Roadmap

Snowflake + Databricks/PySpark — Based on 22,000 Real Job Ads

September 14, 2026 – January 17, 2027

Source: Data with Baraa — *Data Engineer Roadmap: Built From 22,000 Real Job Ads*
Insight: Job ads demand BOTH Snowflake AND Databricks skills. Don't choose — learn both.
Commitment: 25 hours/week
Goal: Job-ready Data Engineer by January 31, 2027

Why Combined?

The video analyzed **22,000 real job ads** across 4,000+ companies. The data shows:

Finding	What It Means for You
Both Snowflake AND Databricks appear in job postings	Learning only one limits your applications by 40–50%
PySpark is the #1 big data skill requested	You need it for Databricks, EMR, and any large-scale processing
dbt is the standard transformation tool	Works on Snowflake AND Databricks — learn once, use everywhere
Two "tech stacks" dominate	Stack A: Snowflake + dbt. Stack B: Databricks + PySpark. You need both.
SQL is non-negotiable	90%+ of job ads mention SQL. It's the foundation of everything.

This roadmap combines both dominant stacks into one learning path.

Table of Contents

- Phase 1: Python + SQL Foundations (Weeks 1–2)
- Phase 2: PySpark Foundations (Weeks 3–4)
- Phase 3: Snowflake + Databricks Side-by-Side (Weeks 5–7)
- Phase 4: dbt — The Universal Transformer (Weeks 8–9)
- Phase 5: Ingestion + Orchestration (Weeks 10–11)
- Phase 6: Infrastructure + Data Quality (Weeks 12–13)
- Phase 7: Streaming (Weeks 14–15)
- Phase 8: Capstone + Job Hunt (Weeks 16–18)

Phase 1: Python + SQL Foundations

Weeks 1–2: September 14 – 27

Theme: You can't build pipelines without speaking the language.

Week 1: Python for Data Engineering

Day	Task	Hours
Mon	`polars` crash course: DataFrames, filters, joins, groupby	4
Tue	`pydantic` for data validation + `requests` for APIs	4
Wed	Build weather fetcher: API → clean → CSV/Parquet	4
Thu	Error handling, logging, `pathlib`, `csv.DictWriter`	4
Fri	Dockerize the script: `Dockerfile`, `docker-compose`	4
Sat	Write `pytest` tests with `unittest.mock`	5
Sun	Push `week-01/` to GitHub	4

Deliverable: Containerized Python script with tests

Week 2: SQL Mastery

Day	Task	Hours
Mon	SELECT, WHERE, JOINs, NULL handling	4
Tue	Aggregation: GROUP BY, HAVING, window functions	4
Wed	CTEs, subqueries, LATERAL joins	4
Thu	Practice: 15 problems on DataLemur / HackerRank	4
Fri	Query optimization: `EXPLAIN`, indexes, execution plans	4
Sat	Build ShopStream dataset, write 10 analytical queries	5
Sun	Push `week-02/sql_notes.md`	4

Deliverable: SQL notes with 10+ solved problems + memory queries

---

Phase 2: PySpark Foundations

Weeks 3–4: September 28 – October 11

Theme: PySpark is the #1 big data skill in job ads. Master it before touching cloud platforms.

Week 3: PySpark Basics

Day	Task	Hours
Mon	Install PySpark, `SparkSession`, RDDs vs DataFrames	4

Day	Task	Hours
Tue	DataFrame ops: `select`, `filter`, `withColumn`, `drop`	4
Wed	Joins: inner, left, broadcast joins	4
Thu	Aggregations: `groupBy`, `agg`, `pivot`	4
Fri	Window functions in PySpark: `Window.partitionBy`	4
Sat	Practice: Load 1M-row dataset, transform, write Parquet	5
Sun	Push `week-03/pyspark_intro.py`	4

Deliverable: PySpark script that reads, transforms, writes Parquet

Week 4: PySpark Advanced + Optimization

Day	Task	Hours
Mon	Spark SQL: `createOrReplaceTempView`, `spark.sql()`	4
Tue	Partitioning: `repartition`, `coalesce`, partition pruning	4
Wed	Caching: `persist`, `cache`, storage levels	4
Thu	Catalyst optimizer, Spark UI, AQE basics	4
Fri	Complex types: arrays, structs, `explode`	4
Sat	Build ETL: 3 CSVs → join → aggregate → partitioned Parquet	5
Sun	Push `week-04/` with optimized Spark job	4

Deliverable: Multi-source PySpark ETL with optimization notes

---

Phase 3: Snowflake + Databricks Side-by-Side

Weeks 5–7: October 12 – November 1

Theme: Job ads ask for BOTH. Learn both on the same dataset.

Week 5: Snowflake Deep Dive

Day	Task	Hours
Mon	Snowflake setup: warehouse, database, TPC-H sample data	4
Tue	Advanced SQL in Snowflake: window functions, CTEs	4
Wed	Time travel, cloning, zero-copy cloning	4
Thu	Query profiling, cost optimization, micro-partitions	4
Fri	Iceberg tables in Snowflake	4

Day	Task	Hours
Sat	Build analytical queries on TPC-H data	5
Sun	Push `week-05/snowflake_notes.md`	4

Deliverable: 5 complex analytical queries in Snowflake worksheet

Week 6: Databricks + Delta Lake

Day	Task	Hours
Mon	Databricks Community Edition: clusters, notebooks, DBFS	4
Tue	Migrate PySpark job to Databricks notebook	4
Wed	Delta Lake: create tables, time travel, `MERGE INTO`	4
Thu	Schema evolution, `OPTIMIZE`, `VACUUM`	4
Fri	Unity Catalog: managed vs external tables	4
Sat	Build medallion architecture: bronze → silver → gold	5
Sun	Push `week-06/databricks_notes.md` + notebooks	4

Deliverable: Bronze + silver + gold Delta tables in Databricks

Week 7: Compare + Contrast

Day	Task	Hours
Mon	Run SAME query on Snowflake vs Databricks SQL	4
Tue	Compare: Iceberg (Snowflake) vs Delta Lake (Databricks)	4
Wed	Data modeling: star schema in BOTH platforms	4
Thu	Cost comparison: Snowflake credits vs Databricks DBUs	4
Fri	Write ADR: "When to use Snowflake vs Databricks"	4
Sat	Build hybrid pipeline: Snowflake for warehouse, Databricks for Spark	5
Sun	Push `week-07/comparison.md`	4

Deliverable: Architecture decision record + hybrid pipeline concept

---

Phase 4: dbt — The Universal Transformer

Weeks 8–9: November 2 – 15

Theme: dbt works on Snowflake AND Databricks. Learn it once, apply everywhere.

Week 8: dbt Core on Snowflake

Day	Task	Hours
Mon	`dbt init`, `profiles.yml`, connect to Snowflake	4
Tue	Staging models, sources, refs	4
Wed	Intermediate + mart models	4
Thu	Tests: `unique`, `not_null`, `relationships`	4
Fri	Jinja macros, `dbt-utils`	4
Sat	Snapshots, documentation, `dbt docs`	5
Sun	Push `week-08/dbt_snowflake/`	4

Deliverable: dbt project on Snowflake with tests + docs

Week 9: dbt on Databricks + Advanced

Day	Task	Hours
Mon	Same dbt project, switch profile to Databricks	4
Tue	Compare: dbt on Snowflake vs dbt on Databricks	4
Wed	dbt Mesh: cross-project references	4
Thu	Slim CI, pre-commit hooks	4
Fri	dbt Cloud vs Core	4
Sat	Build unified dbt project that runs on BOTH	5
Sun	Push `week-09/dbt_unified/`	4

Deliverable: Single dbt project deployable to Snowflake OR Databricks

---

Phase 5: Ingestion + Orchestration

Weeks 10–11: November 16 – 29

Theme: Move data. Schedule pipelines. The operational layer.

Week 10: Airbyte + Ingestion

Day	Task	Hours
Mon	Deploy Airbyte locally, architecture overview	4
Tue	Postgres → Snowflake connection	4
Wed	CDC with Debezium	4
Thu	API ingestion, custom connectors	4

Day	Task	Hours
Fri	dlt (data load tool) comparison	4
Sat	Build ingestion layer for capstone dataset	5
Sun	Push `week-10/ingestion/`	4

Deliverable: 3 sources syncing to Snowflake via Airbyte

Week 11: Apache Airflow

Day	Task	Hours
Mon	Deploy Airflow, first DAGs	4
Tue	TaskFlow API, sensors, task groups	4
Wed	Airflow + dbt (Cosmos)	4
Thu	Retries, SLAs, Slack alerts	4
Fri	Backfills, catchup	4
Sat	Build DAG: Airbyte → dbt → email alert	5
Sun	Push `week-11/dags/`	4

Deliverable: Production DAG orchestrating ingestion → transformation

---

Phase 6: Infrastructure + Data Quality

Weeks 12–13: November 30 – December 13

Theme: Provision as code. Trust your data.

Week 12: Terraform + Cloud

Day	Task	Hours
Mon	Terraform basics: HCL, providers, state	4
Tue	AWS: S3, IAM roles	4
Wed	Snowflake provider: warehouse, database, roles	4
Thu	Databricks provider: clusters, jobs	4
Fri	GitHub Actions + Terraform CI/CD	4
Sat	Provision full stack: S3 + IAM + Snowflake + Databricks config	5
Sun	Push `week-12/terraform/`	4

Deliverable: Terraform modules for both platforms

Week 13: Great Expectations + Observability

Day	Task	Hours
Mon	GX: expectations, validators, checkpoints	4
Tue	GX + dbt integration	4
Wed	GX on Snowflake tables	4
Thu	GX on Delta Lake tables	4
Fri	DataHub / OpenMetadata for lineage	4
Sat	Quality-gated pipeline: dbt → GX → alert	5
Sun	Push `week-13/quality/`	4

Deliverable: Data quality checks on BOTH Snowflake and Databricks

---

Phase 7: Streaming

Weeks 14–15: December 14 – 27

Theme: Real-time data. The skill that separates mid-level from senior.

Week 14: Kafka + Spark Streaming

Day	Task	Hours
Mon	Kafka: topics, partitions, producers, consumers	4
Tue	Python producer/consumer with `confluent-kafka`	4
Wed	Kafka Connect: Debezium CDC	4
Thu	Spark Structured Streaming: `readStream`, `writeStream`	4
Fri	Delta Live Tables: declarative streaming	4
Sat	Build streaming pipeline: Postgres → Kafka → Spark → Delta	5
Sun	Push `week-14/streaming/`	4

Deliverable: Hybrid batch + streaming pipeline

Week 15: CI/CD + Polish

Day	Task	Hours
Mon	GitHub Actions for dbt + Terraform	4
Tue	Pre-commit hooks across all repos	4
Wed	Monitoring: Spark UI, Airflow metrics	4

Day	Task	Hours
Thu	Cost optimization: Snowflake + Databricks	4
Fri	Security: Unity Catalog / Lake Formation basics	4
Sat	Final repo cleanup, READMEs, architecture diagrams	5
Sun	Push everything	4

Deliverable: Production-ready CI/CD across all projects

Phase 8: Capstone + Job Hunt

Weeks 16–18: December 28 – January 17, 2027

Theme: Build the flagship. Get hired.

Week 16: Capstone Design + Build

Day	Task	Hours
Mon	Architecture: Airbyte → Snowflake + Databricks → dbt → Airflow	4
Tue	Terraform all infrastructure	4
Wed	Build ingestion (Airbyte + API + CDC)	4
Thu	Build dbt models (staging → intermediate → mart)	4
Fri	Integrate Airflow DAG	4
Sat	Add GX + alerts	5
Sun	Test end-to-end 5+ times	4

Deliverable: Working end-to-end pipeline on BOTH platforms

Week 17: Polish + Content

Day	Task	Hours
Mon	Add streaming component	4
Tue	Build Streamlit dashboard	4
Wed	Write blog: "Snowflake vs Databricks: A Data Engineer's Guide"	4
Thu	Record 3-minute demo video	4
Fri	Documentation: architecture, runbook, data dictionary	4
Sat	Update LinkedIn, GitHub profile	5
Sun	Resume: 1 page, metrics-focused	4

Deliverable: Professional portfolio + blog + video

Week 18: Apply

Day	Task	Hours
Mon	Apply to 10 jobs	4
Tue	Apply to 10 jobs + follow-ups	4
Wed	Interview prep: SQL (5 problems)	4
Thu	System design: whiteboard your capstone	4
Fri	Behavioral practice	4
Sat	Apply to 10 jobs + networking	5
Sun	Rest + plan January follow-ups	4

Deliverable: 30+ applications sent

The Two Tech Stacks (From the Video)

Stack A: Analytics-First	Stack B: Big Data-First	You Learn Both
Snowflake	Databricks	☑ Week 5–7
dbt	PySpark	☑ Week 3–4, 8–9
SQL-first	Python-first	☑ Week 1–2
Airflow	Airflow	☑ Week 11
Iceberg	Delta Lake	☑ Week 5–7
Great Expectations	Great Expectations	☑ Week 13

Weekly Time Commitment

Day	Hours	Focus
Monday–Friday	3.5–4 hrs/day	New concepts + coding
Saturday	5–6 hrs	Deep work, projects
Sunday	3–4 hrs	Review, polish, push
Weekly Total	**25 hrs**	

Job-Ready Checklist (By January 31, 2027)

- ☐ GitHub: 5+ repos (Python, PySpark, Snowflake, Databricks, dbt, Terraform)
- ☐ Capstone: End-to-end pipeline on BOTH Snowflake and Databricks

- [ ] Blog: "Snowflake vs Databricks: A Data Engineer's Perspective"
- [ ] SQL: 50+ problems solved
- [ ] PySpark: Can write complex DataFrame operations from memory
- [ ] Snowflake: Comfortable with warehouses, Iceberg, cost optimization
- [ ] Databricks: Comfortable with notebooks, Delta Lake, Unity Catalog
- [ ] dbt: Project deployable to both platforms
- [ ] LinkedIn: Updated headline mentioning both stacks
- [ ] Resume: 1 page, metrics-focused, both Snowflake + Databricks listed
- [ ] Demo: 3-minute video
- [ ] Applications: 50+ jobs applied

Certifications to Consider

Certification	Platform	When to Take
Snowflake SnowPro Core	Snowflake	After Week 5
Databricks Data Engineer Associate	Databricks	After Week 6
AWS Cloud Practitioner	AWS	After Week 12

Your First Step — Do This Today

1. Sign up for Databricks Community Edition (free)
2. Sign up for Snowflake free trial (if not done)
3. Install PySpark: `pip install pyspark`
4. Verify: `python -c "from pyspark.sql import SparkSession; print('Ready')"`
5. Create `week-03/pyspark_intro.py` — your first Spark job

Roadmap based on Data with Baraa's analysis of 22,000 real job ads. Combined stack for maximum job market coverage.